The two regular ends of a chiral marking

A recovery of Theorem 1.3 of Leemans–Vandenschrick

We prove that neither $\mathrm{PSL}_3(q)$ nor $\mathrm{PSU}_3(q)$, for $q \geq 2$, is the automorphism group of a chiral polytope of rank greater than five. The acquired classifications of regular subgroups give the two possible local actions. The argument must allow one end to preserve a point and the other a line. This distinction replaces an unjustified inference in the source's even-characteristic proof.

We use the standard group characterization of chiral polytopes, in the notation of Section 2 of the source, with $r \geq 3$. Thus $G = \langle \sigma_1, \ldots, \sigma_{r-1} \rangle$, every σ_i is nonidentity, and $\tau_{ij} = \sigma_i \cdots \sigma_j$ is an involution for $i < j$. For $I \subseteq \{0, \ldots, r-1\}$ put

$$G^I = \langle \tau_{ij} : i \leq j,\ i-1, j \in I \rangle.$$

The actual intersection property is $G^I \cap G^J = G^{I \cap J}$; in particular a singleton index set gives the trivial group. No quotient intersection property will be used.

The regular ends and a proper normal subgroup

Lemma 1. *The involutions $\tau_{12}, \ldots, \tau_{1,r-1}$ form a directly regular string C-marking of rank $r-2$, of type $[|\sigma_3|, \ldots, |\sigma_{r-1}|]$. There is a second such marking $\tau_{r-2,r-1}, \ldots, \tau_{1,r-1}$, with reversed type $[|\sigma_{r-3}|, \ldots, |\sigma_1|]$.*

Proof. For $J \subseteq \{2, \ldots, r-1\}$ the subgroup generated by τ_{1j} for $j \in J$ is exactly $G^{\{0\} \cup J}$. Indeed, each remaining generator with both endpoints in J is $\tau_{i,j} = \tau_{1,i-1}^{-1} \tau_{1,j}$. Actual IP therefore gives all subset intersections of the proposed marking. For $k - j \geq 2$, $\tau_{1j}\tau_{1k} = \tau_{j+1,k}$ is an involution, proving distant commutation. Adjacent products are σ_{j+1}, as asserted.

The even-word subgroup of any involutory marking is generated by its adjacent products. Here it is $\langle \sigma_3, \ldots, \sigma_{r-1} \rangle$, a normal subgroup of index at most two. It cannot contain τ_{12}: its intersection with $G^{\{0,2\}}$ is $G^{\{2\}} = 1$. Thus the index is two. Applying the same argument to the reversed inverse rotation tuple gives the second marking. This also proves the source's Lemma 2.3 directly from its actual intersection convention. □

Lemma 2. *If $r \geq 3$ and some σ_i has order two, then G has a proper nontrivial normal subgroup. Also σ_i and σ_j commute whenever $|i - j| > 2$.*

Proof. First note an elementary consequence of the string relations. If $B^2 = (AB)^2 = (BC)^2 = (ABC)^2 = 1$, then $BAB = A^{-1}$, $BCB = C^{-1}$, and

$$1 = (ABC)^2 = A(BCB)(BAB)C = AC^{-1}A^{-1}C.$$

Taking $A = \sigma_i$, $B = \sigma_{i+1} \cdots \sigma_{j-1}$ and $C = \sigma_j$ proves the distant commutation statement.

Now let $t = \sigma_i$ have order two, and put $L = \langle \sigma_1, \ldots, \sigma_{i-1} \rangle$ and $R = \langle \sigma_{i+1}, \ldots, \sigma_{r-1} \rangle$. Conjugation by t inverts the adjacent generator on either side. It centralizes every longer interval ending immediately before it or starting immediately after it, because both that interval and its product with t are involutions. These intervals and the adjacent generator generate the corresponding side, so t normalizes L, R. Distant commutation shows that the

two sides commute except possibly $\sigma_{i-1}, \sigma_{i+1}$. The identity above, with $B = t$, covers that pair as well. Moreover $L \cap R = 1$ by their disjoint vertex-index sets. Hence $H = LR$ is a normal direct product in G.

Here properness is an actual IP assertion, not an assumption of independence of rotation generators. If $t = lr$ with $l \in L, r \in R$, then

$$r \in \langle L, t \rangle \cap R = G^{\{0,\dots,i\}} \cap G^{\{i,\dots,r-1\}} = 1.$$

It follows that $t \in L \cap \langle t \rangle = G^{\{i-1\}} = 1$, a contradiction. The same calculation holds at an endpoint, with the empty side trivial. Since $r \geq 3$, at least one other nonidentity rotation belongs to H, so H is nontrivial. □

We acquire the complete even-characteristic regular subgroup theorem: a nondegenerate regular string marking in $\mathrm{PSL}_3(F)$ has rank at most four, and at rank four it has, up to coordinates and duality,

$$R_0 = I + E_{23}, \quad R_1 = I + aE_{31}, \quad R_2 = I + bE_{13}, \quad R_3 = I + E_{21}, \qquad ab \neq 0. \tag{1}$$

We also acquire the complete odd-characteristic theorem. Its regular rank bound is four without a nondegeneracy assumption. Its nondegenerate directly regular rank-four markings are the Heisenberg marking $[p, 2p, p]$ and the simplex marking $[3, 3, 3]$ in characteristic five. The other two conic markings have simple groups and are not directly regular. These are actual marked-group statements, not just group orders.

The classical simplicity and order formulas for the projective linear and unitary groups imply that $\mathrm{PSL}_3(q)$ is simple for $q \geq 2$, and $\mathrm{PSU}_3(q)$ is simple for $q \geq 3$. Also $\mathrm{PSU}_3(q)$ embeds in $\mathrm{PSL}_3(q^2)$: the intersection of the scalar centre of $\mathrm{SL}_3(q^2)$ with $\mathrm{SU}_3(q)$ is precisely the latter's centre. Lemmas 1 and 2, and the acquired rank bounds, consequently give $r \leq 6$ for these simple groups.

The small unitary group

Lemma 3. *In* $\mathrm{PSU}_3(2)$ *no two distinct involutions commute. In particular it cannot be the group of a chiral polytope of rank at least five.*

Proof. The scalar kernel has odd order three. Every projective involution therefore has a unique involutory lift, and projectively commuting involutory lifts commute: their scalar commutator has square one. In characteristic two a nonidentity involution of dimension three is $T = I + uf$, where $u \neq 0$, $f \neq 0$ and $f(u) = 0$. Let h be the nonsingular Hermitian form, linear in its second argument. For $x \in \ker f$, preservation of $h(x, y)$ by T gives $h(x, u)f(y) = 0$ for every y. Thus $\ker f = u^{\perp}$, and $f(x) = \lambda h(u, x)$ for a nonzero scalar λ. The equation $f(u) = 0$ makes u isotropic; preservation of h now reduces to $\lambda + \lambda^2 = 0$. Over $\mathbb{F}_4$ it has just one nonzero solution. Consequently each projective centre supports exactly one nonidentity unitary elation.

Two distinct commuting elations in a projective plane have the same centre or the same axis, by the acquired rank-one commutation lemma. Here the axis is the Hermitian polar of the centre, so either condition forces the same centre and hence the same involution. This proves the first claim. A rank-three string C-marking would contain distinct commuting first and third generators. Lemma 1 therefore rules out chiral rank at least five. We have not used, or re-proved, the source's stronger historical computation of no chiral markings at any rank. □

The mixed point and line case

Assume henceforth that a simple group above has a chiral marking of rank six. Define the actual regular ends

$$R = (\tau_{12}, \tau_{13}, \tau_{14}, \tau_{15}), \qquad S = (\tau_{45}, \tau_{35}, \tau_{25}, \tau_{15}). \tag{2}$$

Both are nondegenerate by Lemma 2. In even characteristic normalize R as in (1), using a single duality if necessary. The first, second and last marks of S are then

$$S_0 = R_1 R_3 = I + (e_2 + a e_3) e_1^t, \quad S_1 = R_0 R_3 = I + e_2 (e_1^t + e_3^t), \quad S_3 = R_3. \tag{3}$$

The marks S_0, S_3 have the same axis and different centres. The acquired rank-four normal form says that in its point-preserving orientation the outer pair has the same centre; in the dual orientation it has the same axis. Thus S is in the latter orientation. Its first and third marks share a centre, as do its second and fourth. Writing $u = e_2 + a e_3$, we obtain

$$S_2 = I + uf, \qquad f \neq 0, \qquad f(u) = 0. \tag{4}$$

These conclusions concern the actual axes and centres in (3); they do not infer a common fixed point of the two ends.

Since $\sigma_1 = R_3 S_2$ and $\sigma_5 = R_2 R_3$, the displayed lifts are

$$\Sigma_1 = I + E_{21} + uf, \qquad \Sigma_5 = I + b E_{13} + E_{21}. \tag{5}$$

Lemma 2 requires their projective commutation. The second lift is unipotent. If $\Sigma_1 \Sigma_5 = \zeta \Sigma_5 \Sigma_1$ for a scalar ζ, conjugacy of Σ_5 and $\zeta \Sigma_5$ forces $\zeta = 1$ by their eigenvalues. Thus the lifts themselves commute. But the first row of $\Sigma_5 \Sigma_1 - \Sigma_1 \Sigma_5$ is abf, which is nonzero. This excludes even characteristic.

The two odd-characteristic possibilities

The two ends in (2) have a common first type entry $|\sigma_3|$. If one is a simplex marking, the ambient characteristic is five, and its entry three differs from the Heisenberg entry five. Thus either both are simplex markings or both are Heisenberg markings.

In the simplex case all five σ_i have order three. The classical rotation presentation of the Coxeter group $[3,3,3,3,3]$ identifies its rotation group with A_7. Hence G is a nontrivial quotient of the simple group A_7 and has order 2520. This is impossible here: $|\operatorname{PSL}_3(2)| = 168$, $|\operatorname{PSL}_3(3)| = 5616$, and for $q \geq 4$

$$|\operatorname{PSL}_3(q)| = \frac{q^3(q^3-1)(q^2-1)}{(3, q-1)} \geq 20160.$$

Similarly $|\operatorname{PSU}_3(2)| = 72$, $|\operatorname{PSU}_3(3)| = 6048$, and for $q \geq 4$ its order is at least $4^3(4^3+1)(4^2-1)/3 = 20800$.

Finally suppose both ends are Heisenberg markings. Up to coordinates and a single duality, the acquired normal form gives

$$R_0 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \quad R_1 = \begin{pmatrix} 1 & 0 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix},$$

$$R_2 = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 1 & 1 \end{pmatrix}, \quad R_3 = \operatorname{diag}(-1, -1, 1).$$

These are the unique involutory determinant-one lifts. Therefore

$$S_0 = \begin{pmatrix} -1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \quad S_1 = \operatorname{diag}(-1, 1, -1), \quad S_3 = R_3.$$

The first two marks of S have common centre e_2 and distinct axes. In the displayed Heisenberg normal form the first pair has a common axis and the last pair a common centre; in its dual these roles reverse. Consequently the last pair of S has common axis $x_3 = 0$. This accounts for the dual orientation explicitly, and yields

$$S_2 = \begin{pmatrix} -1 & 0 & u \\ 0 & -1 & v \\ 0 & 0 & 1 \end{pmatrix}.$$

Commutation with S_0 gives $u = 2v$, although the remaining contradiction does not need this extra restriction. We have

$$\Sigma_1 = R_3 S_2 = I - uE_{13} - vE_{23}, \qquad \Sigma_5 = R_2 R_3 = I - E_{32}.$$

Again projective distant commutation lifts to ordinary commutation because Σ_5 is unipotent. Its matrix difference is

$$\Sigma_1 \Sigma_5 - \Sigma_5 \Sigma_1 = uE_{12} + vE_{22} - vE_{33}.$$

Thus $u = v = 0$, making σ_1 trivial, a contradiction. Together with Lemma 3 this completes the claimed rank bound.

Attribution and exact scope

The nonexistence result is Theorem 1.3 of the fixed 2022 source, and Lemmas 1 and 2 reprove its Lemmas 2.3,6.1,6.2. The actual regular subgroup classifications are separately acquired proofs of Section 5. The even argument treats both point and line orientations; the odd argument justifies the homology-axis inference and accounts for the projective scalar in distant commutation. The source's simplex phrase "$q = 5$" is read as characteristic five for this subgroup alternative. The rank bound includes $\mathrm{PSU}_3(2)$ by the elementary unitary argument above, without importing an unspecified computer search. The broader rank-three subgroup classifications, the source's small negative chiral censuses, and its remaining unitary existence questions are separate obligations, not consequences claimed here.